

Isomorphic Graphs:-

Let G_1 and G_2 be the two graphs. The graph G_1 is said to be isomorphic to the graph G_2 is said to be 1 to 1 corresponding b/w the vertex sets and edge sets. In such a way that if e_1 is an edge b/w the 2 end vertices v_1 and v_2 in G_1 , then the corresponding edge f_1 in G_2 with the end vertices u_1 and u_2 in G_2 such a pair of correspondence is called a graph isomorphism.

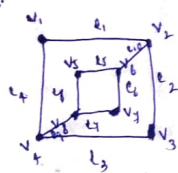
Notes

① If G_1 is isomorphic to G_2 , then
 $\rightarrow$ Both G_1 and G_2 contains same number of vertices. i.e.
 $|V_1| = |V_2|$

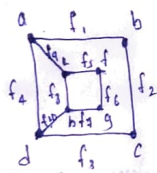
$\rightarrow$ Both G_1 and G_2 contains same number of edges. i.e.
 $|E_1| = |E_2|$

$\rightarrow G_1$ and G_2 preserves adjacency.

Ex:-



(G_1)



(G_2)

$$V = \{8, 8\}$$

$$E = \{10, 10\}$$

$$d(v_1) = 2$$

$$d(v_2) = 3$$

$$d(v_3) = 3$$

$$d(v_4) = 3$$

$$d(u_1) = 3$$

$$d(u_2) = 2$$

$$d(u_3) = 2$$

$$d(u_4) = 3$$

In above graph G_1 and G_2 number of vertices are 8 and edges are 10. And there are 4 vertices of degree 2 and 4 vertices of degree 3.

Now

$v_1 \in G_1$ when there is a corresponding vertex $u_1 \in G_2$
 b, c, g, f

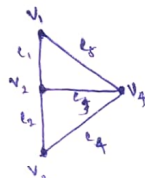
Here v_1 is adjacent to two vertices v_2 and v_4 of degree 3 in G_1 .

In G_2 there is no vertex which is adjacent to the vertices of degree 3

$\therefore$ Adjacency is not preserved

$\therefore$ The graphs are not isomorphic.

Ex:-



(G_1)

$$V_1 = 4$$

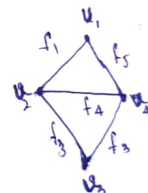
$$E_1 = 5$$

$$d(v_1) = 3$$

$$d(v_2) = 3$$

$$d(v_3) = 3$$

$$d(v_4) = 3$$



(G_2)

$$V_2 = 4$$

$$E_2 = 5$$

$$d(u_1) = 3$$

$$d(u_2) = 3$$

$$d(u_3) = 3$$

$$d(u_4) = 3$$

$\therefore$ Here v_1 corresponds to u_1 ($v_1 \leftrightarrow u_1$)

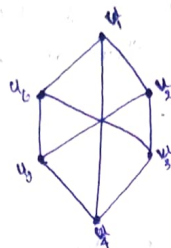
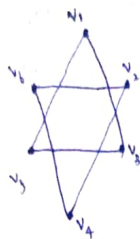
$$v_2 \leftrightarrow u_2$$

$$v_3 \leftrightarrow u_3$$

$$v_4 \leftrightarrow u_4$$

$\therefore$ Adjacency is preserved
 $\therefore$ The graphs are isomorphic.

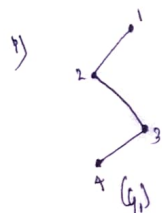
Ex-



$$V_1 = 6$$

$$U_1 = 6$$

Q) Verify the following graphs are isomorphic or not justify your answer.



Sol: In the above graphs G_1, G_2 there are 4 vertices and 5 edges each

and also in G_1 , there are 2 vertices of degree 2 and two vertices of degree 1

$$\text{Here } 1 \in G_1 \Rightarrow f(1) \in G_2 \Rightarrow f(1) = a \in G_2$$

$$4 \in G_1 \Rightarrow f(4) = c \in G_2$$

$$2 \in G_1 \Rightarrow f(2) = b \in G_2$$

$$3 \in G_1 \Rightarrow f(3) = d \in G_2$$

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$$\text{Also } (1,2) \in G_1 \Rightarrow (f(1), f(2)) \in G_2$$

$$(a, b) \in G_2$$

$$(2,3) \in G_1 \Rightarrow (f(2), f(3)) \in G_2$$

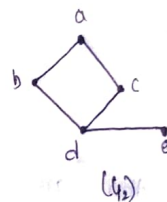
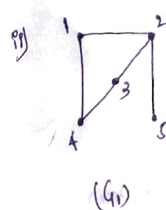
$$(b, d) \in G_2$$

$$(3,4) \in G_1 \Rightarrow (f(3), f(4)) \in G_2$$

$$(d, c) \in G_2$$

∴ It preserves adjacency. or G_1 and G_2 preserves adjacency
also there is one-one correspondence b/w vertices and edges

$$\therefore G_1 \cong G_2$$



Sol: In the above graphs G_1, G_2 , there are 5 vertices and 5 edges each

Also both G_1 and G_2 contains one vertex of degree 3
Three vertices of degree 2 and one vertex of degree 1

$$\text{Here } 1 \in G_1 \Rightarrow f(1) \in G_2 \Rightarrow f(1) = b \in G_2$$

$$2 \in G_1 \Rightarrow f(2) = d \Rightarrow f(2) = d \in G_2$$

$$3 \in G_1 \Rightarrow f(3) = c \Rightarrow f(3) = c \in G_2$$

$$4 \in G_1 \Rightarrow f(4) = a \Rightarrow f(4) = a \in G_2$$

$$5 \in G_1 \Rightarrow f(5) = e \Rightarrow f(5) = e \in G_2$$

if we take $f(1)$ as
a we won't get
correct matching
adjacency

Also

$$(1,2) \in G_1 \Rightarrow (f(1), f(2)) \in G_2$$

$$(b, d) \in G_2$$

$$(2,3) \in G_1 \Rightarrow (f(2), f(3)) = (d, c) \in G_2$$

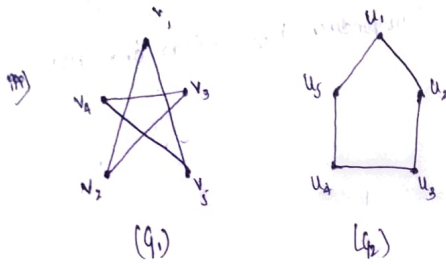
$$(3,4) \in G_1 \Rightarrow (f(3), f(4)) = (c, a) \in G_2$$

$$(4,1) \in G_1 \Rightarrow (f(4), f(1)) = (a, b) \in G_2$$

$$(1,5) \in G_1 \Rightarrow (f(1), f(5)) = (b, d) \in G_2$$

$\therefore G_1, G_2$ preserve isomorphic adjacency also there is one-one correspondence b/w vertices & edges

$$G_1 \cong G_2$$



In above graphs G_1 and G_2 there are 5 vertices and 5 edges each. Also, in both there are 2 degree for all vertices.

$$\text{Here } v_1 \in G_1 \Rightarrow f(v_1) = u_1 \in G_2$$

$$v_2 \in G_1 \Rightarrow f(v_2) = u_2 \in G_2$$

$$v_3 \in G_1 \Rightarrow f(v_3) = u_3 \in G_2$$

$$v_4 \in G_1 \Rightarrow f(v_4) = u_4 \in G_2$$

$$v_5 \in G_1 \Rightarrow f(v_5) = u_5 \in G_2$$

$$\text{Also } (1,2) \in G_1 \Rightarrow (f(1), f(2)) = (u_1, u_2) \in G_2$$

$$(2,3) \in G_1 \Rightarrow (f(2), f(3)) = (u_2, u_3) \in G_2$$

$$(v_3, v_4) \in G_1 \Rightarrow (f(v_3), f(v_4)) = (u_3, u_4) \in G_2$$

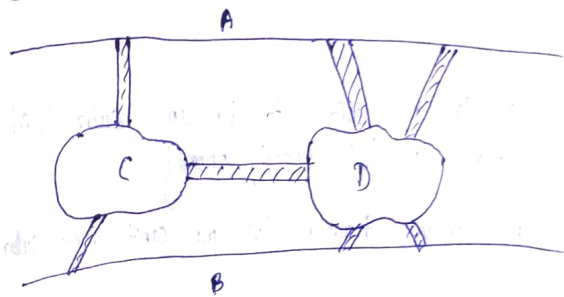
$$(v_4, v_5) \in G_1 \Rightarrow (f(v_4), f(v_5)) = (u_4, u_5) \in G_2$$

$(v_5, v_1) \in G_1 \Rightarrow (f(v_5), f(v_1)) = (u_5, u_1) \in G_2$
 $\therefore G_1, G_2$ preserve adjacency also there is one-one

correspondence b/w vertices & edges

$$G_1 \cong G_2$$

**** Königsberg Bridge problem



Two islands C and D are formed by the river in the present city of Königsberg were connected to each other and to the banks of the river A and B with 7 bridges as shown in the figure.

The problem is to start at any one of the A land areas and walk over each of these 7 bridges exactly once and return to the starting point without swimming across the river.

Euler represented this problem by means of a graph G as shown below: